

On the Minimum Length of Functional Batch Codes with Small Recovery Sets

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Abstract

Batch codes are of potential use for load balancing and private information retrieval in distributed data storage systems. Recently, a special case of batch codes, termed functional batch codes, was proposed in the literature. In functional batch codes, users may query linear combinations of the information symbols, and not only the information symbols themselves, as is the case for standard batch codes. In this work, we consider linear functional batch codes with the additional property that every query is answered by using only a small number of coded symbols. We derive bounds on the minimum length of such codes, and evaluate the bounds numerically.

Keywords: functional batch codes, generating functions, lower bounds.

1 Introduction

Batch codes were first proposed in [1] for load balancing in multi-server distributed data storage systems. They are also potentially useful in applications such as network switches and private information retrieval. Linear batch codes [2] are a special case of batch codes, where the encoding procedure is a linear transformation applied to the information symbols over a finite field. The properties of batch codes were extensively studied over the years: the reader can refer, for example, to [3, 4, 5, 6, 7, 8]. Batch codes with a small size of recovery sets were studied in [9].

Functional batch codes, a special case of batch codes, were studied in [10, 11, 12, 13, 14]. Unlike regular batch codes, where only information

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symbols can be queried, in functional batch codes any linear combination of information symbols can be queried and retrieved. In [14], a lower bound on the minimal length of a functional batch code was derived.

Both linear batch codes and linear functional batch codes can be defined by their generator matrices $\mathbf{G}$. The length of the code is, therefore, equal to the number of columns in $\mathbf{G}$. In the literature, usually, the number of code symbols used for a single retrieval is unrestricted. However, in practice, this could lead to extremely high traffic in the system. Therefore, it would be beneficial to restrict the number of symbols used in the retrieval of each query. In this work, we investigate such functional batch codes, where the number of symbols used in each retrieval is small, typically bounded from above by a small constant. In particular, we investigate the asymptotic behavior of lower bounds on the minimal length of such codes.

2 Preliminaries and notation

In this section, we present notations and some results that will be used in the sequel. We denote by $\mathbb{N}$ the set of natural numbers (including zero), $\mathbb{N}^+ = \mathbb{N} \setminus \{0\}$. We also denote by $\mathbb{Z}$ the set of integers, and by $\mathbb{R}$ the set of real numbers. We adopt the notation $[n]$ for $n \in \mathbb{N}$, $n \geq 1$, to denote the set $\{1, 2, \dots, n\}$. Throughout the paper, let $\mathbb{F}_2$ be a field with two elements. For a matrix $\mathbf{G}$, we denote by $\mathbf{g}^{(j)}$ its j -th column. Additionally, $\mathbf{e}_i$ denotes the unit vector with a one in position i and zeros elsewhere, and $\mathbf{0}$ denotes the all-zero vector. For a set T of vectors over $\mathbb{F}_2$, we denote by $\langle T \rangle$ its linear span.

Definition 1. For $n \in \mathbb{N}$ and $m \in \mathbb{Z}$ with $0 \leq m \leq n$, we define $(n)_m := n(n-1) \cdots (n-m+1)$; in particular, $(n)_0 = 1$ and $(n)_n = n!$.

Numbers $(n)_m$ are often called *falling factorials*. We use the following well-known result.

Lemma 1. Let $m, n \in \mathbb{N}^+$, $n \geq m$. It holds: $(n)_m \leq (n-(m-1)/2)^m \leq n^m$.

Proof. The first inequality follows from the arithmetic mean-geometric mean inequality; the second one is obvious. $\square$

Definition 2. Let $i_1 + \dots + i_m = n$, where $i_j \in \mathbb{N}$ for $j \in [m]$, and $n \in \mathbb{N}$. A multinomial coefficient is defined as follows:

$$\binom{n}{i_1, i_2, \dots, i_m} := \frac{n!}{i_1! i_2! \dots i_m!}.$$

We sometimes write this as $\binom{n}{\mathbf{i}}$, where $\mathbf{i} = (i_1, i_2, \dots, i_m)$.

Note that the multinomial coefficient above in fact counts the number of partitions of $[n]$ into m disjoint (labeled) subsets of $[n]$ of sizes $i_1, i_2, \dots, i_m$, respectively, for $\sum_{j=1}^m i_j = n$.

Theorem 2. For arbitrary numbers $x_1, x_2, \dots, x_m \in \mathbb{R}$ and for $n \in \mathbb{N}$, the following identity holds:

$$(x_1 + x_2 + \dots + x_m)^n = \sum_{\substack{i_1 + i_2 + \dots + i_m = n \\ i_1, i_2, \dots, i_m \geq 0}} \binom{n}{i_1, i_2, \dots, i_m} x_1^{i_1} x_2^{i_2} \dots x_m^{i_m}.$$

Consider a (finite or infinite) power series $S(x) = s_0 + s_1x + \dots + s_nx^n + \dots$. We denote by $[x^n]S(x)$ the coefficient s_n (corresponding to the term x^n) in this series.

Next, we introduce batch codes. Assume that there is a k -symbol database, which is to be distributed among n devices. A client would like to retrieve an arbitrary subset (or *batch*) of t symbols by reading the data stored on the devices. If too many reading requests are directed to the same device, that could create a heavy load on the device, thus slowing down the performance of the system. Batch codes could prevent that situation by improved load balancing. In this work, we study optimal trade-offs between the number of devices n and the number of queries t . We focus on binary linear batch codes.

Definition 3 ([1, 7]). An $[n, k, t]$ linear batch code $\mathcal{C}$ over $\mathbb{F}_2$ encodes a vector $\mathbf{x} \in \mathbb{F}_2^k$ into a vector $\mathbf{y} \in \mathbb{F}_2^n$, $\mathbf{y} = (y_1, y_2, \dots, y_n)$, as $\mathbf{y} = \mathbf{x} \cdot \mathbf{G}$, such that for any t -tuple of queries $(x_{i_1}, x_{i_2}, \dots, x_{i_t})$, $i_\ell \in [k]$ for $\ell \in [t]$, there exist mutually disjoint subsets $R_{i_1}, R_{i_2}, \dots, R_{i_t}$ of $[n]$, such that each of the values of x_{i_ℓ} , $\ell \in [t]$, can be retrieved only from the symbols of $\mathbf{y}$ with indices in R_{i_ℓ} , respectively. The $k \times n$ matrix $\mathbf{G}$ over $\mathbb{F}_2$ is called the *generator matrix* of $\mathcal{C}$.

The rate of $\mathcal{C}$ is defined as $r := k/n$. For simplicity, an $[n, k, t]$ linear batch code is sometimes referred to as a t -batch code. We remark that each encoded symbol y_j , $j \in [n]$, can be written as $y_j = \sum_{i=1}^k g_{i,j}x_i = \mathbf{x} \cdot \mathbf{g}^{(j)}$, where

$$\mathbf{G} = \left(\mathbf{g}^{(1)} \mid \mathbf{g}^{(2)} \mid \dots \mid \mathbf{g}^{(n)} \right) = (g_{i,j})_{i \in [k], j \in [n]}.$$

The following definition naturally generalizes Definition 3.

Definition 4. An $[n, k, t]$ linear *functional* batch code $\mathcal{C}$ over $\mathbb{F}_2$ encodes a vector $\mathbf{x} \in \mathbb{F}_2^k$ into a vector $\mathbf{y} \in \mathbb{F}_2^n$ as $\mathbf{y} = \mathbf{x} \cdot \mathbf{G}$, such that for any $(\alpha_1, \alpha_2, \dots, \alpha_t)$, $\alpha_\ell \in \mathbb{F}_2^k$ for $\ell \in [t]$, there exist mutually disjoint subsets $R_1, R_2, \dots, R_t$ of $[n]$, and for any $\ell \in [t]$, the value $\mathbf{x} \cdot \alpha_\ell^T$ can be retrieved only from the symbols of $\mathbf{y}$ with indices in R_ℓ , respectively.

For $\ell \in [t]$, we denote $\alpha_\ell = (\alpha_{1,\ell}, \alpha_{2,\ell}, \dots, \alpha_{k,\ell})$. We say that the functional query is

$$(\mathbf{x} \cdot \alpha_1^T, \mathbf{x} \cdot \alpha_2^T, \dots, \mathbf{x} \cdot \alpha_t^T) = \left(\sum_{i=1}^k \alpha_{i,1}x_i, \sum_{i=1}^k \alpha_{i,2}x_i, \dots, \sum_{i=1}^k \alpha_{i,t}x_i \right),$$

or simply $(\alpha_1, \alpha_2, \dots, \alpha_t)$. We assume $\alpha_\ell \neq \mathbf{0}$ for all ℓ , otherwise the retrieval is trivial.

We note that Definition 3 corresponds to the case where $\alpha_1, \alpha_2, \dots, \alpha_t$ are all unit vectors. It is straightforward to see that an $[n, k, t]$ linear functional batch code is in particular an $[n, k, t]$ linear batch code. Since we consider only linear codes in this work, we drop the word “linear” in the sequel.

Theorem 1 in [2] characterizes the recoverability of the batch of queries in a batch code. Below, we generalize it towards functional batch codes.

Theorem 3. Let $\mathcal{C}$ be an $[n, k, t]$ functional batch code. It is possible to retrieve $(\alpha_1, \alpha_2, \dots, \alpha_t)$ simultaneously if and only if there exist t mutually-disjoint subsets $R_1, R_2, \dots, R_t$ of indices in $[n]$, and for each $R_\ell, \ell \in [t]$, there exists a linear combination of columns of $\mathbf{G}$ indexed by that set, which is equal to the column vector α_ℓ^T .

Proof. First, assume that there exist t non-intersecting subsets $R_1, R_2, \dots, R_t$ of indices in $[n]$, and for each $R_\ell, \alpha_\ell^T \in \langle \mathbf{g}^{(j)} \rangle_{j \in R_\ell}$. For each $\ell \in [t]$, take

$$\alpha_\ell^T = \sum_{j \in R_\ell} \gamma_j \cdot \mathbf{g}^{(j)},$$

where all $\gamma_j \in \mathbb{F}_2$. Due to linearity, the encoding of $\mathbf{x} = (x_1, x_2, \dots, x_k)$ can be written as $\mathbf{y} = \mathbf{x} \cdot \mathbf{G}$. Then,

$$\mathbf{x} \cdot \alpha_\ell^T = \mathbf{x} \cdot \left(\sum_{j \in R_\ell} \gamma_j \cdot \mathbf{g}^{(j)} \right) = \sum_{j \in R_\ell} \gamma_j \cdot (\mathbf{x} \cdot \mathbf{g}^{(j)}) = \sum_{j \in R_\ell} \gamma_j \cdot y_j.$$

Therefore, the value of $\mathbf{x} \cdot \alpha_\ell^T$ can be obtained by querying only the values of y_j for $j \in R_\ell$. The conclusion follows from the fact that all R_ℓ are disjoint.

Next, we show the opposite direction of Theorem 3. Let R_ℓ , for $\ell \in [t]$, be a set of indices of entries in $\mathbf{y}$, which are used to retrieve $\mathbf{x} \cdot \alpha_\ell^T$. We show that $\alpha_\ell^T \in \langle \mathbf{g}^{(j)} \rangle_{j \in R_\ell}$.

Denote the vector space $W_\ell := \langle \mathbf{g}^{(j)} \rangle_{j \in R_\ell}$. Assume for a contradiction that $\alpha_\ell \notin W_\ell$. Recall that the dual space of W_ℓ , denoted by $W_\ell^\perp$, consists of all the vectors orthogonal to any vector in W_ℓ . Since $\alpha_\ell^T \notin (W_\ell^\perp)^\perp = W_\ell$, there exists a vector $\mathbf{x}^T \in W_\ell^\perp$, which is not orthogonal to α_ℓ^T , i.e. $\mathbf{x} \cdot \alpha_\ell^T \neq 0$. On the other hand, this vector $\mathbf{x}^T$ is orthogonal to any vector $\mathbf{g}^{(j)}$ for $j \in R_\ell$.

Consider the encodings of the vectors $\mathbf{x}$ and $\mathbf{0}$, $\mathbf{x} \cdot \mathbf{G}$ and $\mathbf{0} \cdot \mathbf{G}$, respectively. In both cases, all the coordinates of $\mathbf{y}$ indexed by R_ℓ are zeros. Therefore, the result of the retrieval of α_ℓ in both cases is the same, yet $\mathbf{x} \cdot \alpha_\ell^T \neq \mathbf{x} \cdot \mathbf{0}^T$. We obtain a contradiction. $\square$

Definition 5. Let $\mathbf{G} = (\mathbf{g}^{(1)} \mid \mathbf{g}^{(2)} \mid \dots \mid \mathbf{g}^{(n)})$ be a $k \times n$ matrix over $\mathbb{F}_2$. A *recovery set* for a (nonzero) vector $\alpha \in \mathbb{F}_2^k \setminus \{\mathbf{0}\}$ is a subset $R \subseteq [n]$ of

column indices such that α is contained in the span $\langle \mathbf{g}^{(j)} \rangle_{j \in R}$ of the columns of $\mathbf{G}$ with indices in R . We say R is a *minimal recovery set* for α if R does not contain a proper subset that is also a recovery set for α .

Definition 6. Let $\mathcal{C}$ be an $[n, k, t]$ (functional) batch code. We say that $\mathcal{C}$ has *locality* $r \in \mathbb{N}^+$, if for any batch $(\alpha_1, \alpha_2, \dots, \alpha_t)$ there exist disjoint recovery sets $R_1, R_2, \dots, R_t$ such that $|R_\ell| \leq r$ for all ℓ . Such a code will also be denoted as an $[n, k, t, r]$ (functional) batch code.

Example 1. Consider the following linear binary code $\mathcal{C}$ whose binary 2×3 generator matrix is given by

$$\mathbf{G} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

The encoding is given by $(y_1, y_2, y_3) = \mathbf{x}\mathbf{G} = (x_1, x_2, x_1 + x_2)$, where $\mathbf{x} = (x_1, x_2) \in \mathbb{F}_2^2$.

We can retrieve any pair (α_1, α_2) as shown in Table 1. Therefore, $\mathcal{C}$ is a $[3, 2, 2, 2]$ functional batch code.

α_1, α_2	$\mathbf{x} \cdot \alpha_1^T$	$\mathbf{x} \cdot \alpha_2^T$	R_1, R_2
$(1, 0), (1, 0)$	$x_1 = y_1$	$x_1 = y_2 + y_3$	$\{1\}, \{2, 3\}$
$(1, 0), (0, 1)$	$x_1 = y_1$	$x_2 = y_2$	$\{1\}, \{2\}$
$(1, 0), (1, 1)$	$x_1 = y_1$	$x_1 + x_2 = y_3$	$\{1\}, \{3\}$
$(0, 1), (1, 0)$	$x_2 = y_2$	$x_1 = y_1$	$\{2\}, \{1\}$
$(0, 1), (0, 1)$	$x_2 = y_1 + y_3$	$x_2 = y_2$	$\{1, 3\}, \{2\}$
$(0, 1), (1, 1)$	$x_2 = y_2$	$x_1 + x_2 = y_3$	$\{2\}, \{3\}$
$(1, 1), (1, 0)$	$x_1 + x_2 = y_3$	$x_1 = y_1$	$\{3\}, \{1\}$
$(1, 1), (0, 1)$	$x_1 + x_2 = y_3$	$x_2 = y_2$	$\{3\}, \{2\}$
$(1, 1), (1, 1)$	$x_1 + x_2 = y_1 + y_2$	$x_1 + x_2 = y_3$	$\{1, 2\}, \{3\}$

Table 1: Recovery sets for possible queries α_1, α_2 .

Definition 7. Let $k \in \mathbb{N}$, $k \geq 1$. A binary $[2^k - 1, k]$ *simplex code* is a linear code of length $n = 2^k - 1$ and dimension k over $\mathbb{F}_2$, whose binary $k \times n$ generator matrix $\mathbf{G}$ contains all nonzero vectors in $\mathbb{F}_2^k$ as columns exactly once.

It is conjectured in [10] that an $[2^k - 1, k]$ simplex code is a $[2^k - 1, k, 2^{k-1}, 2]$ functional batch code for any $k \geq 1$ (for example, for $k = 2$ we obtain the $[3, 2, 2, 2]$ functional batch code in Example 1). Moreover, it is shown in [5] that if $\mathbf{G}$ is a generator matrix of a $[2^k - 1, k]$ simplex code, then the double-simplex code with the generator matrix $[\mathbf{G} \mid \mathbf{G}]$ is a $[2^{k+1} - 2, k, 2^k, 2]$ functional batch code.

Prior results

In [14], the authors presented lower bounds on the minimum length n of an $[n, k, t]$ functional batch code.

Theorem 4. [14, Theorem 23] Suppose that $\mathcal{C}$ is an $[n, k, t]$ functional batch code over $\mathbb{F}_2$. Then, $n/k \geq t/\log(t+1) + o(1)$.

In our work, we build on some ideas from the proof of Theorem 23 in [14]. We recall the main idea of the proof. Assume that the disjoint subsets $R_1, R_2, \dots, R_t \subseteq [n]$ are used for recovery of t functional queries. With each such set R_ℓ , we associate a labeling of the indices in $[n]$ by labels ℓ , $\ell \in [t]$, where index j is assigned a label ℓ if $j \in R_\ell$, and it is assigned a special label 0 if index j does not occur in any of the aforementioned recovery sets. The number of such labellings is $(t+1)^n$. We remark that distinct queries have distinct batches of minimal recovery sets, and thus distinct labellings.

The number of possible ordered batches of queries $(\alpha_1, \alpha_2, \dots, \alpha_t)$ for $\alpha_i \in \mathbb{F}_2^k \setminus \{\mathbf{0}\}$, $i \in [t]$, equals $(2^k - 1)^t$. Since the number of distinct labellings is bounded from below by the number of batches of queries, we obtain that $(t+1)^n \geq (2^k - 1)^t$. $\square$

We remark that in the proof of Theorem 23 in [14], it was assumed that the queries are distinct, hence the number of ways to choose t queries is $\binom{2^k - 1}{t} t!$. In our proof, we allow the same queries to be repeated more than once. In that case, there are $(2^k - 1)^t$ different ordered batches of queries. That approach allows for a slightly more accurate estimate on the number of batches of queries, when compared with the proof in [14].

3 Lower bounds on the minimal length

In this section, we employ an approach based on generating functions in order to obtain lower bounds on the minimum length of $[n, k, t, r]$ functional batch codes. Consider an arbitrary batch of t queries $\alpha_1, \alpha_2, \dots, \alpha_t$. Denote the corresponding pairwise-disjoint minimal recovery sets by $R_1, R_2, \dots, R_t \subseteq [n]$. Assume that the entries of $\mathbf{y}$ are indexed by the elements of $[n]$. Similarly to the approach in [14], we assign some label $\ell \in \{0, 1, \dots, t\}$ to each entry index j in $[n]$. Label $\ell \in [t]$ is assigned to entry $j \in [n]$ if the corresponding y_j is used in the recovery set R_ℓ , $|R_\ell| \leq r$. If entry j is not used in any recovery set, then it is assigned the label 0.

Let $\theta_{t,r}(n)$ denote the number of distinct labellings of disjoint subsets $R_1, \dots, R_t$ of $[n]$ of size at most r , or, equivalently, the number of distinct labellings of the numbers $[n]$ with labels from $\{0, 1, \dots, t\}$, using each nonzero label at least once and at most r times. Our derivations of bounds will be based on the following observation in [14].

Theorem 5. If $\mathcal{C}$ is an $[n, k, t, r]$ functional batch code, then we have $\theta_{t,r}(n) \geq (2^k - 1)^t$.

For the proof of Theorem 5, see the discussion immediately after Theorem 4.

Obviously, we have that

$$\theta_{t,r}(n) = \sum_{\substack{i_0 \in \mathbb{N}, i_1, i_2, \dots, i_t \in [r] : \\ i_0 + i_1 + i_2 + \dots + i_t = n}} \binom{n}{i_0, i_1, \dots, i_t}. \quad (1)$$

Let

$$\Theta_{t,r}(x) := \sum_{n=1}^{\infty} \frac{\theta_{t,r}(n)}{n!} x^n$$

denote the exponential generating function for the sequence $(\theta_{t,r}(n))_{n \in \mathbb{N}}$. Write

$$G_r(x) = \frac{1}{1!}x + \frac{1}{2!}x^2 + \dots + \frac{1}{r!}x^r,$$

and note that $e^x = \sum_{j=0}^{\infty} \frac{x^j}{j!}$. By using (1), we find that

$$\Theta_{t,r}(x) = e^x \cdot (G_r(x))^t. \quad (2)$$

We can use (2) to derive a useful recursion for the values of $\theta_{t,r}(n)$. Indeed, since

$$\Theta_{t,r}(x) = e^x G_r(x)^t = G_r(x) \Theta_{t-1,r}(x) = (x + x^2/2 + \dots + x^r/r!) \Theta_{t-1,r}(x),$$

we have

$$\begin{aligned} \theta_{t,r}(n) &= n! \cdot [x^n] \left(\left(\frac{x}{1!} + \frac{x^2}{2!} + \dots + \frac{x^r}{r!} \right) \Theta_{t-1,r}(x) \right) \\ &= n! \left(\frac{\theta_{t-1,r}(n-1)}{(n-1)! \cdot 1!} + \frac{\theta_{t-1,r}(n-2)}{(n-2)! \cdot 2!} + \dots + \frac{\theta_{t-1,r}(n-r)}{(n-r)! \cdot r!} \right) \\ &= \binom{n}{1} \theta_{t-1,r}(n-1) + \binom{n}{2} \theta_{t-1,r}(n-2) + \dots + \binom{n}{r} \theta_{t-1,r}(n-r). \end{aligned} \quad (3)$$

Note that, in addition, $\theta_{0,r}(n) = 1$ for all $n \geq 1$. Recursion (3) provides a convenient way to compute a table of the values of $\theta_{t,r}(n)$. Moreover, since

$$\theta_{t-1,r}(n-1) \geq \theta_{t-1,r}(n-2) \geq \dots \geq \theta_{t-1,r}(n-r),$$

we obtain that for $n \geq 2r-1$:

$$\theta_{t,r}(n) \leq \left(\binom{n}{1} + \dots + \binom{n}{r} \right) \cdot \theta_{t-1,r}(n-1) \leq r \cdot \binom{n}{r} \cdot \theta_{t-1,r}(n-1), \quad (4)$$

which can be useful to prove bounds on $\theta_{t,r}(n)$ by induction. We now use (2) to derive alternative expressions for the values of $\theta_{t,r}(n)$ that are easier to analyze.

Example 2. By using (2), for $r = 2$ we have:

$$\begin{aligned}\theta_{t,2}(n) &= n! \cdot [x^n] \left((x + x^2/2)^t \sum_{j=0}^{\infty} \frac{x^j}{j!} \right) = n! \cdot \sum_{i=0}^{\min\{t, n-t\}} \frac{\binom{t}{i}}{2^i (n-t-i)!} \\ &\leq \frac{(n)_t}{2^t} \cdot \sum_{i=0}^t \binom{t}{i} 2^{t-i} (n-t)_i \leq (n)_t \cdot (n-t+2)^t / 2^t, \quad (5)\end{aligned}$$

where in the last step we used Lemma 1. $\square$

We now derive an expression for $\theta_{t,r}(n)$ for general r , by extending the approach in Example 2. Below, if $\mathbf{i} = (i_1, \dots, i_r)$, then we write $|\mathbf{i}| := i_1 + 2i_2 + \dots + ri_r$; note that $t \leq |\mathbf{i}| \leq rt$. By using (2), we have

$$\begin{aligned}\theta_{t,r}(n) &= n! \cdot [x^n] \left((x + x^2/2 + \dots + x^r/r!)^t \sum_{j=0}^{\infty} \frac{x^j}{j!} \right) \\ &= n! \cdot \sum_{\mathbf{i}} \frac{\binom{t}{\mathbf{i}}}{(1!)^{i_1} (2!)^{i_2} \dots (r!)^{i_r} (n-|\mathbf{i}|)!} \\ &= (n)_t \cdot \sum_{\mathbf{i}} \frac{\binom{t}{\mathbf{i}} (n-t)_{|\mathbf{i}|-t}}{(1!)^{i_1} (2!)^{i_2} \dots (r!)^{i_r}}, \quad (6)\end{aligned}$$

where the sum is over all r -tuples $\mathbf{i} = (i_1, i_2, \dots, i_r)$ of non-negative integers with $i_1 + i_2 + \dots + i_r = t$ and $|\mathbf{i}| \leq n$. By Lemma 1, we have $(n-t)_{|\mathbf{i}|-t} \leq (n-t)^{|\mathbf{i}|-t}$, thus

$$\theta_{t,r}(n) \leq (n)_t \cdot \sum_{\mathbf{i}} \binom{t}{\mathbf{i}} \left(\frac{(n-t)}{1!} \right)^{i_1} \left(\frac{(n-t)^2}{2!} \right)^{i_2} \dots \left(\frac{(n-t)^r}{r!} \right)^{i_r} (n-t)^{-t}$$

$$\begin{aligned}&\leq (n)_t \left(\frac{(n-t)^1}{1!} + \frac{(n-t)^2}{2!} + \dots + \frac{(n-t)^r}{r!} \right)^t (n-t)^{-t} \\ &= (n)_t \left(\frac{1}{1!} + \frac{(n-t)}{2!} + \dots + \frac{(n-t)^{r-1}}{r!} \right)^t \quad (7)\end{aligned}$$

$$\leq (n - (t-1)/2)^t (1 + (n-t)/2! + \dots + (n-t)^{r-1}/r!)^t, \quad (8)$$

where in the last step we used Lemma 1. Finally, if $n \geq t+r$, then $1 \leq (n-t)/2! \leq \dots \leq (n-t)^{r-1}/r!$, hence

$$\theta_{t,r}(n) \leq \left((n - (t-1)/2) \cdot \frac{(n-t)^{r-1}}{(r-1)!} \right)^t. \quad (9)$$

We use our results to obtain the following lower bounds on n .

Theorem 6. Suppose that $\mathcal{C}$ is an $[n, k, t, r]$ functional batch code, $n \geq t+r$. Then,

$$2^k - 1 \leq (n - (t-1)/2)(n-t)^{r-1}/(r-1)!. \quad (10)$$

Proof. We use Theorem 5 together with (9) to obtain (10). Note that (9) holds for $n \geq t + r$. $\square$

Corollary 1. If an $[n, k, t, r]$ binary functional batch code exists and $n \geq t + r$, then

$$n \geq t - (t + 1)/2r + \left((2^k - 1)(r - 1)! \right)^{1/r}.$$

Proof. We use the result in Theorem 6. By the arithmetic mean-geometric mean inequality, we have that

$$(n - (t - 1)/2)(n - t)^{r-1} \leq \left(n - t + \frac{t + 1}{2r} \right)^r.$$

We can use the recursion (4) to obtain another (similar) bound. $\square$

Theorem 7. If an $[n, k, t, r]$ binary functional batch code exists and if $n \geq \max\{t + 1, 2r - 1\}$, then

$$n \geq (t + r)/2 - 1 + \left((2^k - 1)(r - 1)! \right)^{1/r}.$$

Proof. By using (4) t times, and noting that $\theta_{0,r}(n - t) = 1$ if $n \geq t + 1$, we obtain

$$\theta_{t,r}(n) \leq \frac{r \cdot (n)_r}{r!} \cdot \theta_{t-1,r}(n - 1) \leq \dots \leq r^t \cdot (n)_r \cdot (n - 1)_r \cdot (n - t + 1)_r / (r!)^t.$$

Now, using the arithmetic mean-geometric mean inequality, we find that

$$\begin{aligned} rt((n)_r(n - 1)_r \dots (n - t + 1)_r)^{1/(rt)} &\leq \sum_{i=0}^{t-1} \sum_{j=0}^{r-1} (n - i - j) \\ &= rtn - r \sum_{i=0}^{t-1} i - t \sum_{j=0}^{r-1} j = rtn - rt(t - 1)/2 - tr(r - 1)/2, \end{aligned}$$

hence

$$\theta_{t,r}(n) \leq \frac{(n - (t + r)/2 + 1)^{rt}}{((r - 1)!)^t}.$$

The bound in the theorem follows from Theorem 5. $\square$

When $r = 2$, the inequality (5) and Theorem 5 imply

$$\begin{aligned} (2^k - 1) &\leq (n)_t^{1/t} (n - t + 2)/2 \leq (n - (t - 1)/2)(n - (t - 2))/2 \\ &\leq \frac{1}{2}(n - 3t/4 + 5/4)^2, \end{aligned} \tag{11}$$

where the second transition follows from Lemma 1, and the third transition follows from the arithmetic mean-geometric mean inequality. The following result follows from (11).

Theorem 8. For any $[n, k, t, 2]$ functional batch code, $t \geq 1$, we have

$$n \geq \sqrt{2(2^k - 1)} + 3t/4 - 5/4.$$

4 Numerical results

In this section, we present numerical results for the bounds derived. In Table 2, we compare lower bounds on the minimal length with the upper bound implied by the double-simplex code, for $r = 2$. In columns 3-5, we show the lower bound in Theorem 8, the lower bound implied by the exact computation of $\theta_{t,r}(n)$ and substitution into Theorem 5, and the upper bound implied by the analysis in [5], respectively. $\sim$

k	t	Theorem 8	Exact $\theta_{t,r}(n)$	Construction [5]
2	4	5	5	6
3	8	9	10	14
4	16	17	19	30
5	32	31	38	62
6	64	58	74	126
7	128	111	146	254

Table 2: Comparison of lower and upper bounds on minimal n for $r = 2$.

Table 3 shows the values of the lower bound on n in Theorem 7 for various values of k, t, r , and compares them to the bound [14, Theorem 23] (when the $o(1)$ term is ignored).

k	[14, Theorem 23], $t = 2$	$t = 2, r = 2$	$t = 2, r = 3$	$t = 3, r = 3$	$t = 2, r = 5$
5	7	7	6	6	-
6	8	9	7	8	-
7	9	13	8	9	9
8	11	17	10	10	10
9	12	24	12	13	10
10	13	33	15	15	11
11	14	47	18	18	12
12	16	65	22	23	13
13	17	92	27	28	14
14	18	129	34	34	16
15	19	183	42	43	18

Table 3: Lower bounds in [14, Theorem 23] and Theorem 7.

5 Conclusions

In this work, we considered functional batch codes with a restricted size of recovery sets. By using exponential generating functions, we obtained estimates on the number of labellings of the indices of codeword symbols. Based on those estimates, we derived bounds on the minimal length of a code.

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